



WORKSHEET 4F

QUESTION 1

- Suppose the register AX = 5, BX = 6
- Swap the numbers of ax and bx so that ax gets 6 and bx gets 5. Use the concept of Stack.
- Push and Pop instructions must use.

PUSH AX
PUSH BX
POP AX
POP BX

QUESTION 2

Suppose that AX= 1234h, BX= 5678h, CX = 9ABCh, and SP = 100h.
Write the contents of AX, BX, CX, and SP after executing the following instructions:

PUSH AX
PUSH BX
XCHG AX,CX
POP CX
PUSH AX
POP BX

Initial State:

- AX = 1234h
- BX = 5678h
- CX = 9ABCh
- SP = 0100h

After Instructions:

Instruction	AX	BX	CX	SP
Initial	1234	5678	9ABC	0100
PUSH AX	1234	5678	9ABC	00FE
PUSH BX	1234	5678	9ABC	00FC
XCHG AX,CX	9ABC	5678	1234	00FC
POP CX	9ABC	5678	5678	00FE
PUSH AX	9ABC	5678	5678	00FC
POP BX	9ABC	9ABC	5678	00FE



QUESTION 3

Read in three characters 123 and output them in reverse order.
Hints, use the concept of push and pop.

```
MOV AH, 01h
INT 21h
PUSH AX      ; '1'
```

```
MOV AH, 01h
INT 21h
PUSH AX      ; '2'
```

```
MOV AH, 01h
INT 21h
PUSH AX      ; '3'
```

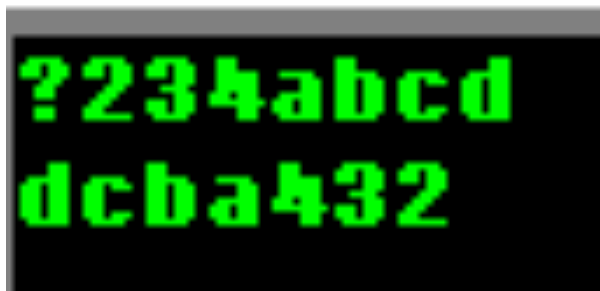
```
POP DX
MOV AH, 02h
INT 21h
```

```
POP DX
MOV AH, 02h
INT 21h
```

```
POP DX
MOV AH, 02h
INT 21h
```

QUESTION 4

Take a string from user. Once user hits enters reverse the string given by the user. Must take input from user. Hints: Use the concept of push and pop. Use procedures





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```
.MODEL SMALL  
.STACK 100h  
.DATA  
.CODE
```

```
READ_STRING PROC  
READ:  
    MOV AH, 01h  
    INT 21h  
    CMP AL, 0Dh  
    JE DONE  
    PUSH AX  
    JMP READ  
DONE:  
    RET  
READ_STRING ENDP
```

```
PRINT_REVERSE PROC  
PRINT:  
    CMP SP, 0100h  
    JE EXIT  
    POP DX  
    MOV AH, 02h  
    INT 21h  
    JMP PRINT  
EXIT:  
    RET  
PRINT_REVERSE ENDP
```

```
MAIN PROC  
    MOV AX, @DATA  
    MOV DS, AX  
  
    CALL READ_STRING  
    CALL PRINT_REVERSE  
  
    MOV AH, 4Ch  
    INT 21h  
MAIN ENDP
```



QUESTION 5

Suppose AX contains, 1 and BX contains FFFFh. For MUL what the value of Dx=?

- A. 1 **B. 0** C. none D. all

QUESTION 6

Example 9.1: Suppose AX contains, 1 and BX contains FFFFh

Instructions	Decimal Product	Hex Product	DX	AX	CF/OF
MUL BX	65535	0000FFFF	0000	FFFF	0
IMUL BX	-1	FFFFFFFF	FFFF	FFFF	0

- For **MUL**, **DX = 0**, so CF/OF= **0**.
- for **IMUL**, the signed interpretation of BX is **-1**, and the product is also **-1**. In 32 bits, this is FFFFFFFFh. CF/OF = 0 because DX is the sign extension of AX.
-

QUESTION 7

Example 9.2 Suppose AX contains FFFFh and BX contains FFFFh:

Instruction	Decimal product	Hex product	DX	AX	CF/OF
MUL BX	4294836225	FFFE0001	FFFE	0001	1
IMUL BX	1	00000001	0000	0001	0

For MUL, CF/OF = 1 because DX is not 0. This reflects the fact that the product FFFE0001h is too big to fit in AX.

For IMUL, AX and BX both contain -1, so the product is 1. DX has the sign extension of AX, so CF/OF = 0.